

GNSS AI Translation Engine

One-Page Executive Summary — Precision-GNSS.com Automated Technical Translation (EN→ES)

Status: FASE 4 & 5 complete (60 tests, real-API validated) · FASE 0-3, 6-10 pending staging access

An independent software system that translates Precision-GNSS.com's technical GNSS/RTK content from English into European Spanish using the DeepSeek API, built to protect technical accuracy — exact numbers, units, conditions and certainty — rather than treat translation as a generic text-rewrite task. Every translation passes through three independent AI checks before it is ever eligible for publication.

Source Text → **Translator** (DeepSeek) → **Technical Reviewer** (DeepSeek) → **Terminology Validator** (DeepSeek) → Draft in WordPress (never auto-published)

Technology

Python 3.10 · DeepSeek API (deepseek-v4-pro / deepseek-v4-flash, OpenAI-compatible, JSON mode) · Pydantic v2 for strict response validation · pytest (60 automated tests) · designed integration layer: WordPress REST API + a minimal custom mu-plugin (gnss-bridge) to reach WPML's PHP-only translation-linking hooks safely · SQLite for translation memory.

Status

Area	Status
Translation Engine — Translator, Reviewer, Terminology Validator, chunking	Complete
Glossary Engine	Complete (seed glossary)
Translation Memory & QA scoring	Not started
WordPress/WPML connector, extraction & orchestration	Blocked — needs staging access

Cost (DeepSeek, empirically measured on 4 real site pages)

Using the real chunk_text() splitter and measured system-prompt sizes against 4 live precision-gnss.com pages (avg. 2,371 words): ~\$0.0064 per page translation-only, ~\$0.0129 per page with full independent review (both on deepseek-v4-pro) — a flat ~\$0.00085 per 1,000 characters across every page tested. Using deepseek-v4-flash for the QA calls cuts the full-pipeline cost to ~\$0.0085/page with no loss of translation quality.

Bottom line: well under one cent per page for full quality-assured translation. A 50-page site translation pass costs roughly \$0.64 total.

Next Steps

- Blocked on staging access: WordPress connector, Elementor/Yoast extraction, WPML orchestration.
- Available now: QA Engine (numeric/unit/URL checks), Translation Memory, gnss-bridge skeleton.